

**Seventh Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ERROR CORRECTING CODES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) What is meant by LRC and VRC with respect to error detecting codes ?
Give example of the same. 3(CO1)
- (b) A DMS transmits messages x_1, x_2, x_3 with probabilities 0.3, 0.4, 0.3 respectively. The Channel Matrix is shown below. Determine Input Entropy, Output Entropy, Joint Entropy and Conditional Entropy.

$$P(y/x) = \begin{vmatrix} 0.8 & 0.2 & 0 \\ 0 & 1 & 0 \\ 0 & 0.3 & 0.7 \end{vmatrix} \quad 7(\text{CO1})$$

2. The generator matrix (6, 3) for Linear Block Code is given by –

$$G = \begin{vmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{vmatrix}$$

- (i) Find parity check matrix.
- (ii) List all code vectors.
- (iii) Comment on error corrective capability.
- (iv) Decode the received codewords $Y_1 = 110 \ 111, Y_2 = 011 \ 101$.
10(CO2,3)

3. A message 10 11 01 is to be transmitted in cyclic code with a Generator polynomial $G(p) = p^4 + p^3 + 1$. Obtain the transmitted code word. How many check bits does the encoded message contain ? 10(CO2,3)

4. A Convolution Code is described by $V_1 = [1 \ 1 \ 1]$; $V_2 = [1 \ 0 \ 1]$;
 Draw the convolution encoder.
 The received sequence for above encoder 010010.
 Decode the sequence using Sequential decoding Algorithm. 10(CO5)

5. Explain in brief Recursive and Non Recursive Turbo Codes with an example. 10(CO4)

6. Discuss in detail with example :
 - (i) RS Codes.
 - (ii) LDPC Codes. 10(CO4)

